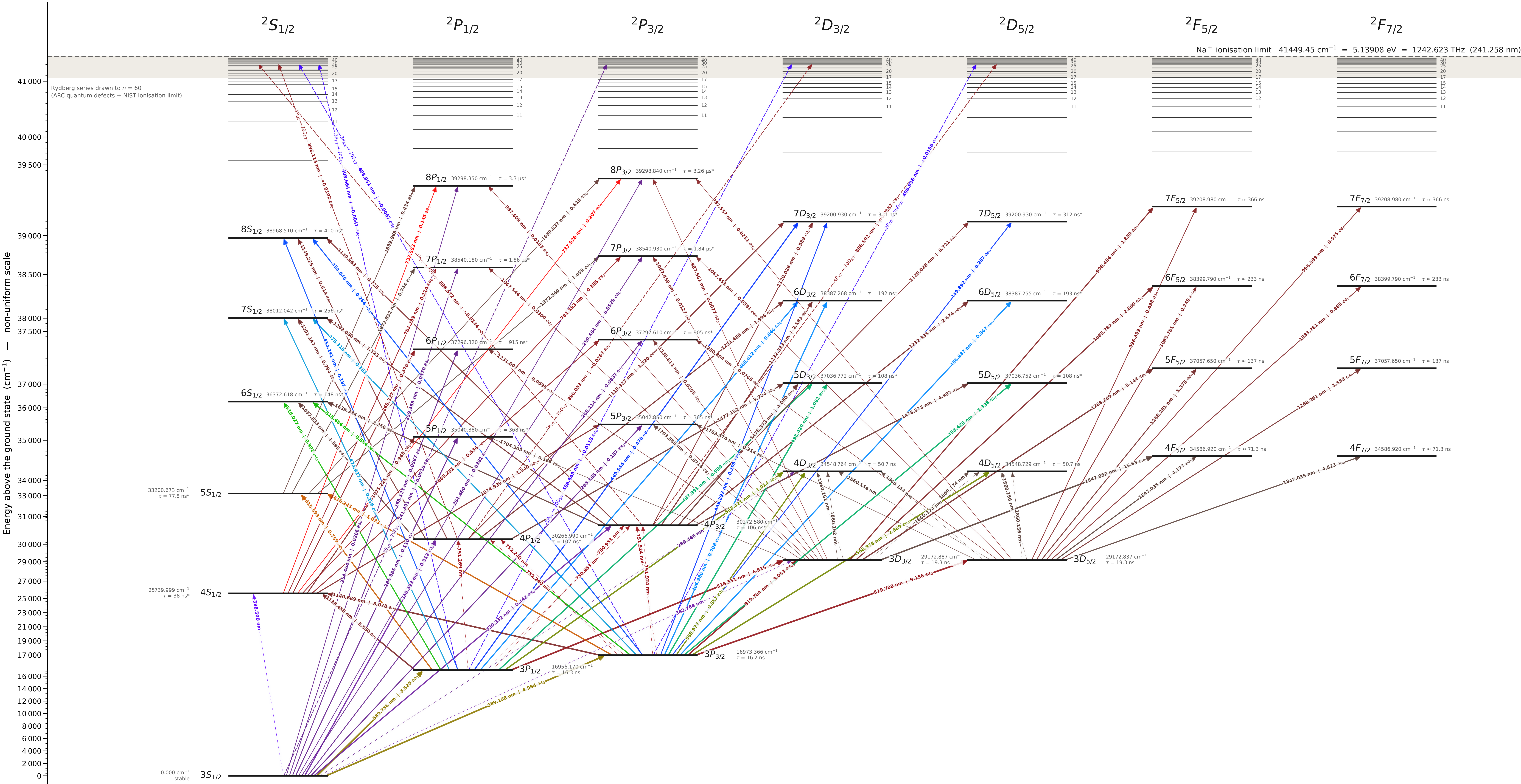
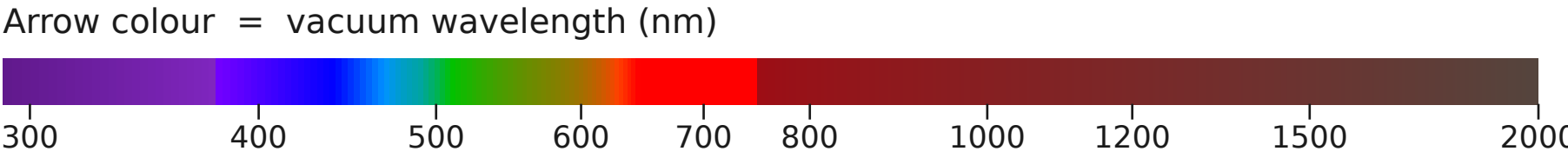


# Sodium-23

<sup>23</sup>Na (Na I), nuclear spin  $I = 3/2$ . Energy levels and transitions below 2  $\mu\text{m}$ : 36 levels, 106 lines.

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle J||er||J' \rangle$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked  $\approx$ ).



## Key transitions

| Transition                      | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $\tilde{\nu}$ ( $\text{cm}^{-1}$ ) | $\langle J  er  J' \rangle$ ( $e a_0$ ) | A ( $\text{s}^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Typical use                                  |
|---------------------------------|--------------------|--------------------|-------------|------------------------------------|-----------------------------------------|-----------------------|-------------------------|------------|----------------------------------------------|
| $3S_{1/2} \rightarrow 3P_{3/2}$ | 589.1583           | 588.9951           | 508.8487    | 16973.366                          | 4.984                                   | 6.152e+07             | 9.795                   | 100.0      | D2 line                                      |
| $3S_{1/2} \rightarrow 3P_{1/2}$ | 589.7558           | 589.5924           | 508.3332    | 16956.170                          | 3.525                                   | 6.135e+07             | 9.765                   | 100.0      | D1 line                                      |
| $3P_{3/2} \rightarrow 3D_{5/2}$ | 819.7077           | 819.4824           | 365.7309    | 12199.471                          | 9.156                                   | 5.140e+07             | 8.259                   | 100.0      |                                              |
| $3P_{1/2} \rightarrow 3D_{3/2}$ | 818.5505           | 818.3255           | 366.2480    | 12216.717                          | 6.815                                   | 4.290e+07             | 8.259                   | 83.4       |                                              |
| $3P_{3/2} \rightarrow 4S_{1/2}$ | 1140.6888          | 1140.3766          | 262.8170    | 8766.633                           | 5.078                                   | 1.760e+07             | 4.187                   | 66.6       |                                              |
| $3P_{3/2} \rightarrow 4D_{5/2}$ | 568.9783           | 568.8205           | 526.8961    | 17575.363                          | 2.569                                   | 1.210e+07             | 3.139                   | 63.5       |                                              |
| $3P_{1/2} \rightarrow 4D_{3/2}$ | 568.4210           | 568.2633           | 527.4127    | 17592.594                          | 1.914                                   | 1.010e+07             | 3.139                   | 53.0       |                                              |
| $3P_{1/2} \rightarrow 4S_{1/2}$ | 1138.4557          | 1138.1441          | 263.3326    | 8783.829                           | 3.580                                   | 8.800e+06             | 4.187                   | 33.4       |                                              |
| $3P_{3/2} \rightarrow 3D_{3/2}$ | 819.7043           | 819.4790           | 365.7324    | 12199.521                          | 3.053                                   | 8.570e+06             | 8.259                   | 16.6       |                                              |
| $3P_{3/2} \rightarrow 5S_{1/2}$ | 616.2452           | 616.0747           | 486.4824    | 16227.307                          | 1.073                                   | 4.980e+06             | 2.045                   | 38.7       |                                              |
| $3P_{3/2} \rightarrow 5D_{3/2}$ | 498.4199           | 498.2808           | 601.4858    | 20063.406                          | 1.092                                   | 4.880e+06             | 1.472                   | 8.8        |                                              |
| $3P_{3/2} \rightarrow 5D_{5/2}$ | 498.4204           | 498.2813           | 601.4852    | 20063.386                          | 1.338                                   | 4.880e+06             | 1.468                   | 52.9       |                                              |
| $3P_{1/2} \rightarrow 5D_{3/2}$ | 497.9930           | 497.8541           | 602.0013    | 20080.602                          | 0.999                                   | 4.090e+06             | 1.472                   | 44.2       |                                              |
| $3S_{1/2} \rightarrow 4P_{3/2}$ | 330.3319           | 330.2369           | 907.5491    | 30272.580                          | 0.442                                   | 2.750e+06             | 1.501                   | 28.2       | 330 nm UV line: narrow-line cooling / UV MOT |
| $3S_{1/2} \rightarrow 4P_{1/2}$ | 330.3929           | 330.2978           | 907.3815    | 30266.990                          | 0.312                                   | 2.730e+06             | 1.490                   | 28.0       | 330 nm UV line                               |
| $3P_{1/2} \rightarrow 5S_{1/2}$ | 615.5929           | 615.4225           | 486.9979    | 16244.503                          | 0.759                                   | 2.500e+06             | 2.045                   | 19.4       |                                              |
| $3P_{3/2} \rightarrow 6D_{3/2}$ | 466.9864           | 466.8557           | 641.9726    | 21413.902                          | 0.708                                   | 2.490e+06             | 0.827                   | 8.0        |                                              |

$\lambda$  and  $\nu$  from NIST level energies (centre of gravity, no hyperfine structure);  $\lambda$  air uses standard air (Ciddor).  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ . Branch: share of the upper level's decays that takes this line.

## Rydberg excitation wavelengths (vacuum, nm)

| Lower $\rightarrow$ series      | n = 30   | n = 40   | n = 50   | n = 70   | n = 100  | n $\rightarrow$ $\infty$ | d% ARC |
|---------------------------------|----------|----------|----------|----------|----------|--------------------------|--------|
| $3S_{1/2} \rightarrow nP_{3/2}$ | 242.012  | 241.675  | 241.522  | 241.391  | 241.323  | 241.258                  | 0.0011 |
| $3P_{3/2} \rightarrow nS_{1/2}$ | 410.806  | 409.792  | 409.337  | 408.951  | 408.750  | 408.562                  | 0.0067 |
| $3P_{3/2} \rightarrow nD_{5/2}$ | 410.610  | 409.711  | 409.297  | 408.936  | 408.745  | 408.562                  | 0.0158 |
| $4S_{1/2} \rightarrow nP_{3/2}$ | 641.837  | 639.474  | 638.406  | 637.491  | 637.012  | 636.559                  | 0.0044 |
| $3D_{5/2} \rightarrow nP_{3/2}$ | 823.219  | 819.336  | 817.583  | 816.083  | 815.298  | 814.557                  | 0.0003 |
| $3D_{5/2} \rightarrow nF_{7/2}$ | 822.729  | 819.133  | 817.480  | 816.046  | 815.286  | 814.557                  | 0.0457 |
| $4P_{3/2} \rightarrow nS_{1/2}$ | 905.535  | 900.623  | 898.431  | 896.572  | 895.608  | 894.705                  | 0.0144 |
| $4P_{3/2} \rightarrow nD_{5/2}$ | 904.583  | 900.233  | 898.235  | 896.502  | 895.584  | 894.705                  | 0.0357 |
| $5S_{1/2} \rightarrow nP_{3/2}$ | 1231.589 | 1222.918 | 1219.015 | 1215.683 | 1213.944 | 1212.301                 | 0.0096 |
| $4D_{5/2} \rightarrow nP_{3/2}$ | 1476.769 | 1464.320 | 1458.728 | 1453.960 | 1451.472 | 1449.124                 | 0.0023 |
| $4D_{5/2} \rightarrow nF_{7/2}$ | 1475.191 | 1463.672 | 1458.401 | 1453.842 | 1451.432 | 1449.124                 | 0.0847 |
| $4F_{7/2} \rightarrow nD_{5/2}$ | 1483.574 | 1471.910 | 1466.575 | 1461.961 | 1459.523 | 1457.188                 | 0.0087 |
| $5P_{3/2} \rightarrow nS_{1/2}$ | 1594.151 | 1578.993 | 1572.267 | 1566.583 | 1563.642 | 1560.890                 | 0.0244 |
| $5P_{3/2} \rightarrow nD_{5/2}$ | 1591.204 | 1577.794 | 1571.665 | 1566.368 | 1563.569 | 1560.890                 | 0.0612 |
| $6S_{1/2} \rightarrow nP_{3/2}$ | 2021.162 | 1997.914 | 1987.519 | 1978.677 | 1974.073 | 1969.732                 | 0.0167 |

Rydberg term energies from ARC quantum defects + NIST ionisation limit; lower level from NIST. d% =  $\langle J||er||J' \rangle$  to  $n = 70$  in  $e a_0$  (calculated).  
 $S \rightarrow nP$  values come from a near-cancelling integral: treat them as an order of magnitude.

## <sup>23</sup>Na hyperfine structure ( $I = 3/2$ )

| Level      | A (MHz)    | B (MHz)  | Shift of each F level from the centre of gravity (MHz) |
|------------|------------|----------|--------------------------------------------------------|
| $3S_{1/2}$ | 885.813064 |          | - F=1: -1107.27 F=2: +664.36                           |
| $3P_{1/2}$ | 94.390(20) |          | - F=1: -117.99 F=2: +70.79                             |
| $3P_{3/2}$ | 18.532(6)  | 2.722(7) | F=0: -66.09 F=1: -50.28 F=2: -15.94 F=3: +42.38        |
| $4S_{1/2}$ | 203.60(20) |          | - F=1: -254.50 F=2: +152.70                            |
| $3D_{5/2}$ | 0.1085(24) |          | - F=1: -0.57 F=2: -0.35 F=3: -0.03 F=4: +0.41          |
| $3D_{3/2}$ | 0.527(25)  |          | - F=0: -1.98 F=1: -1.45 F=2: -0.40 F=3: +1.19          |
| $4P_{1/2}$ | 30.60(10)  |          | - F=1: -38.25 F=2: +22.95                              |
| $4P_{3/2}$ | 6.02(6)    | 0.97(6)  | F=0: -21.37 F=1: -16.32 F=2: -5.24 F=3: +13.79         |
| $5S_{1/2}$ | 77.40(14)  |          | - F=1: -96.75 F=2: +58.05                              |
| $4D_{5/2}$ | 0.029(6)   |          | - F=1: -0.15 F=2: -0.09 F=3: -0.01 F=4: +0.11          |
| $4D_{3/2}$ | 0.215(15)  |          | - F=0: -0.81 F=1: -0.59 F=2: -0.16 F=3: +0.48          |
| $5P_{1/2}$ | 13.469(4)  |          | - F=1: -16.84 F=2: +10.10                              |
| $5P_{3/2}$ | 2.6360(23) | 0.370(8) | F=0: -9.42 F=1: -7.16 F=2: -2.25 F=3: +6.02            |

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).